

TORSION: A QUICK ENGINEERING GUIDE

For the TorsiLab torsion calculator • solid shafts, hollow shafts, and future spring-torsion analysis

What is torsion?

Torsion is the twisting of a structural member caused by an applied torque (turning moment). In a shaft, torque produces a shear-stress distribution across the cross-section and causes the shaft to rotate. For a circular shaft in the elastic range, shear stress increases approximately linearly from zero at the centre to a maximum at the outer surface. Torsion calculations help engineers check strength, stiffness, and serviceability.

Where is torsion used?

Mechanical engineering	Drive shafts, axles, transmission shafts, couplings, gear shafts, motor shafts and rotating machinery.
Civil / structural	Members and connections subjected to eccentric loading, bridge components, structural tubes and reinforced-concrete members.
Aerospace & automotive	Propeller/rotor shafts, drivetrain components, steering systems, aircraft structures and lightweight tubular members.
Other fields	Robotics, manufacturing, biomechanics, sports equipment, tools, wind-turbine drivetrains and experimental material testing.

Core torsion formulas — circular shafts

Shear stress: $\tau = T\rho / J$	Maximum shear stress: $\tau_{\max} = TR / J$
Angle of twist: $\phi = TL / (GJ)$	Shear strain: $\gamma = \rho\phi / L$
Torsional rigidity: GJ	Torque–twist relation: $T = GJ\phi / L$

Polar second moment of area (J)

Solid circular shaft: $J = \pi d^4 / 32$	Hollow circular shaft: $J = \pi(D^4 - d^4) / 32$
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Variables and units

Symbol	Meaning	Typical SI unit
T	Applied torque / twisting moment	N·m
τ	Shear stress at radius ρ	Pa (N/m²)
$\tau_{\max}$	Maximum shear stress at the outer radius	Pa
ρ	Radial distance from the shaft centre	m
R	Outer radius of the shaft	m
J	Polar second moment of area	m⁴
L	Shaft length	m
G	Shear modulus (modulus of rigidity)	Pa
ϕ	Angle of twist	rad
γ	Shear strain	dimensionless
d / D	Solid diameter / hollow shaft outer diameter (d = inner diameter)	m

Important note on torsion springs

A **torsion spring** is different from a circular shaft even though both involve twisting moments. A helical torsion spring primarily stores energy through bending of its wire, so its detailed stress and deflection equations depend on spring geometry and material. A useful first-order relationship is $T = k\phi$, where k is the spring rate. For spring design, use the appropriate spring geometry, stress-concentration/Wahl factors, operating limits, and material data rather than applying the shaft equation blindly.

Assumptions behind the basic shaft equations

The standard equations above assume a circular, prismatic shaft, elastic and homogeneous material, small deformation, Saint-Venant torsion, and a torque distribution for which the elementary circular-shaft solution applies. For non-circular sections, plastic torsion, warping-dominated members, stress concentrations, or complex load paths, more advanced analysis may be required.

TorsiLab • Interactive Torsion Analysis | Always verify critical designs against the applicable engineering code, material data, and a qualified engineer.